

TITLE: **G-SHIELD** (Gesture controlled Smart Hazard Rover)

TEAM NAME: Mecha-Cytron

TEAM ID: YH610

PS ID: Challenge 9

DOMAIN: Hardware

TEAM LEADER NAME: Deepak S

MEMBER-1: Bino Benedict L

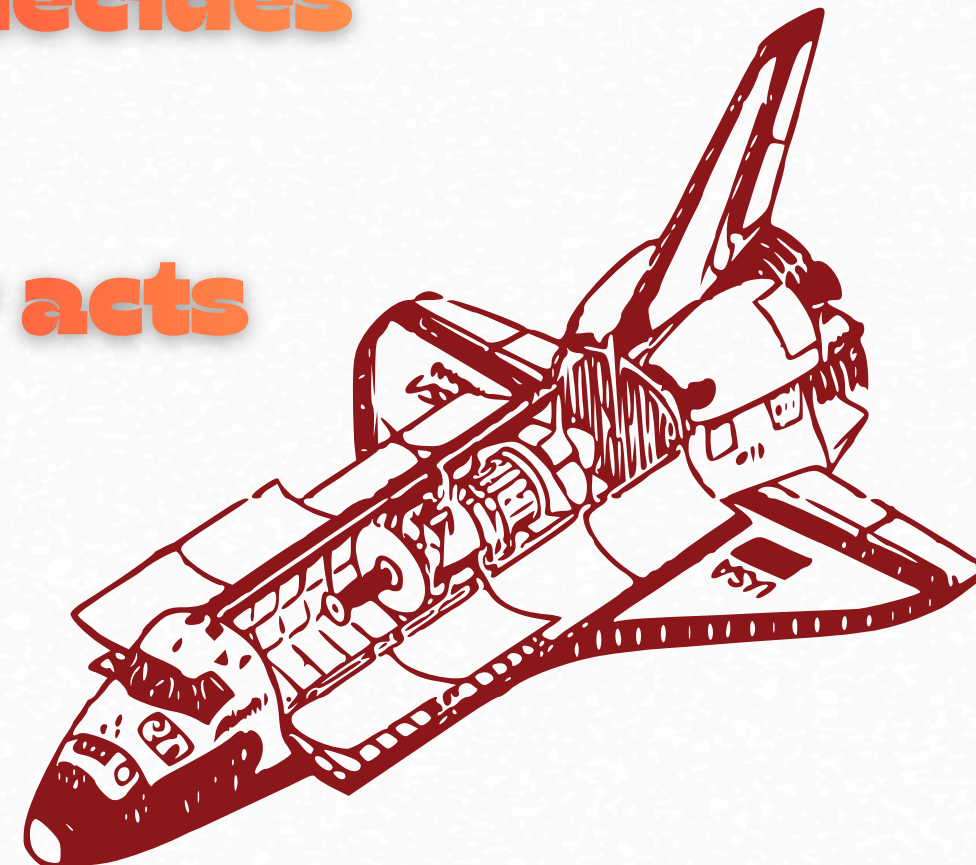
MEMBER-2: Prashant A

MEMBER-3: Srihari Siddharth M

MEMBER-4: Sri Hari G

MEMBER-5: Vishwa Prasath P

Gesture in
↓
Safety decides
↓
Rover acts



PROBLEM STATEMENT

Challenge 9 → Motion To Command

When the environment is dangerous, the operator should not have to enter it

Existing interaction gap

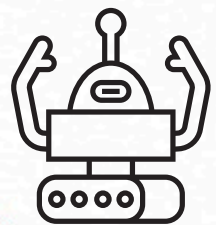
Conventional robotic control relies on physical controllers, keyboards or mice, creating dependence on traditional interfaces.

Hazardous Environment Gap

Remote robots can protect the operator from physical exposure, but blindly executing commands can still create unsafe motion.

Challenge Gap

A practical hands-free system must reliably recognize human gestures, transmit commands wirelessly, respond with low delay and provide immediate feedback.



Actual
Requirement

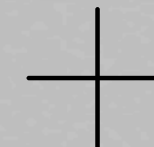


A human optimized wireless system to have control over in hazardous situations

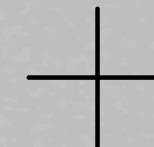
Missing Implementation



Natural Human Control



Wireless Communication



Intellectual Local Safety

PROPOSED SOLUTION

G-SHIELD: A Human-in-the-Loop Safety-Aware Rover

*Convert hand motion into wireless robotic intel and let the rover
validate that intent before actuation.*

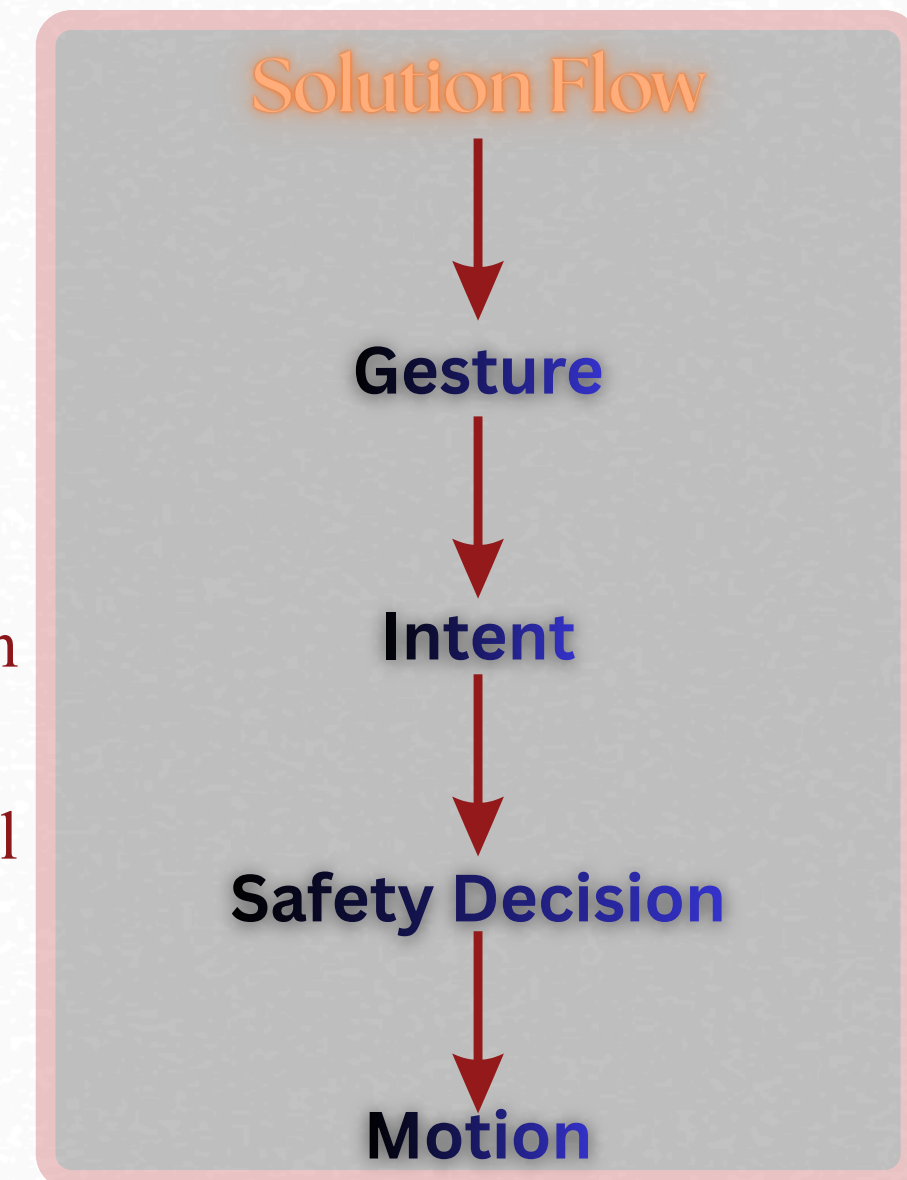
The proposed solution is of to completely eradicate the implementation of human usage in hazardous locations and make use of the advanced robotic with human control and wireless module for the risky tasks.

- The wearable **MPU6050** captures and senses the hand orientation
- **ESP S32** processes motion and recognizes pre-defined hand gestures
- A **wireless command link** is used to transfer the movement command to the rover
- The **Safety Decision Layer** uses the combined rover stability, obstacle state and communication state to protect the rover
- **PWM, BTS7960 and Differential Drive** acts to execute the safe movement of the four wheel platform



CORE DIFFERENTIATOR

The main uniqueness would be that the motor doesn't directly follow the hand gesture commands rather check for the surroundings and verify the movement safety of the rover with the inbuilt sensors



OBJECTIVE

**Our objective is to build a responsive,
reliable and safety-aware hands-free
robotic control system**

**We deliberately prioritize a small number of robust
gestures over a large gesture vocabulary**

01

Reliable gestures

Recognize a small, robust command set with calibration and neutral/dead-zone handling.

02

Responsive wireless link

Transmit compact commands with local processing and a communication-loss fail-safe.

03

Safe execution

Use obstacle sensing to override unsafe movement rather than blindly executing commands.

04

Adaptive motion

Use rover IMU state to reduce PWM on demanding inclines / unstable conditions.

05

Immediate feedback

Return command / rover-state feedback and provide live visual reconnaissance when available.

**TARGET
CONTROL
LOOP**

Gesture → Recognition → Wireless Command → Safety Validation → Motor Response → Feedback

METHODOLOGY

From raw motion to safe physical action

Calibrate

Establish Neutral
Hand Position

Neutral Reference

↓
Personalized Motion
Thresholds

Sense

MPU6050

- Acceleration
- Orientation/Tilt

Classify

Thresholds
+
Dead Zone

Pre-Defined Gestures

↓
Command Stability

Transmit

Compact Wireless
Command Packet

Gestures

↓
Command

↓
Wireless Link

Arbtrirate

- Obstacle + tilt
+ link state
- May allow /
modify / stop

Actuate

PWM

↓
BTS7960

↓
Motor Control

Feedback

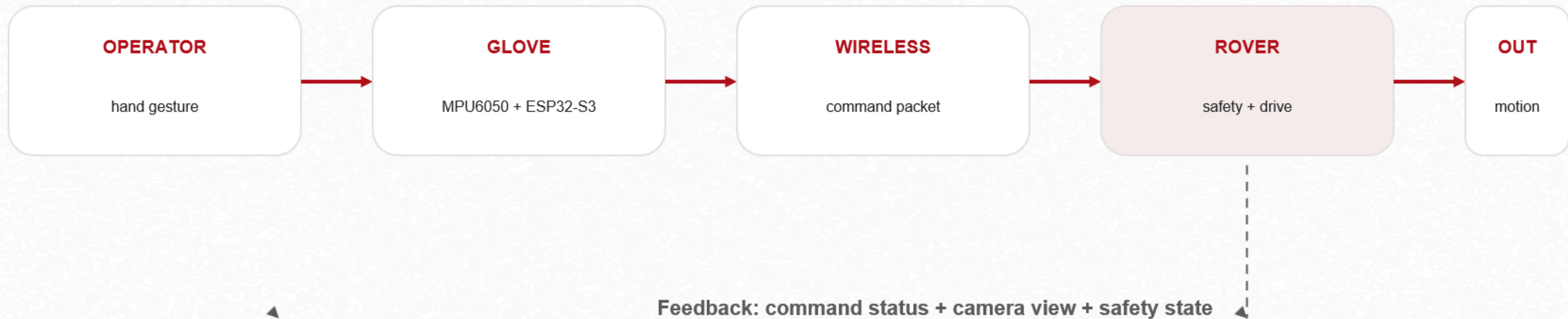
Recognized Gesture

↓
Rover State

↓
Visual Feedback

Gesture	Front Tilt	Backward Tilt	Left Tilt	Right Tilt	Neutral
Command	Forward	Backward	Left turn	Right turn	Stop

WORKFLOW



Decision point

Execute = safe | Modify = adapt PWM | Reject = stop

FAIL-SAFE RULE: UNCERTAINTY OR UNSAFE STATE → STOP

SYSTEM ARCHITECTURE

A THREE-LAYER EMBEDDED CONTROL ARCHITECTURE

WEARABLE INTERFACE

- **MPU6050:**
 - Motion sensing
 - Hand orientation
 - Gesture input
- **ESP32-S31:**
 - Sensor acquisition
 - Calibration
 - Gesture classification
 - Command generation

WIRELESS + DECISION

- **Wireless Link:**
 - Command transmission
 - Low-overhead packet exchange
- **Safety Decision Layer:**
Inputs:
 - Ultrasonic obstacle state
 - Rover IMU state
 - Communication health
Outputs:
 - Allow/Adapt/Stop

ROVER

- **BTS7960 × 2**
 - Left-side drive
 - Right-side drive
- **12V DC Gear Motors × 4**
 - Differential-drive locomotion
- **Ultrasonic × 2**
 - Obstacle sensing
- **MPU6050 × 1**
 - Pitch / roll / stability monitoring
- **Encoders × 4**
 - Wheel-motion feedback / validation
- **ESP32-S CAM**
 - Remote visual reconnaissance

DATA FLOW: SENSOR DATA ⇒ DECISION ⇒ ACTUATION ⇒ FEEDBACK

POWER FLOW: LI-ION BATTERY ⇒ MOTOR DRIVERS / EMBEDDED ELECTRONICS

IMPLEMENTATION

Subsystem	Hardware	Role
Wearable	MPU6050 + ESP32-S3	Motion capture + gesture classification
Rover control	ESP32-S CAM / ESP32 board	Command decoding + control logic
Drive	BTS7960 x2 + 12V gear motors x4	Left/right differential drive
Safety	Ultrasonic x2 + rover MPU6050	Obstacle and stability monitoring
Feedback	ESP32-S CAM	Remote camera / visual survey
Telemetry	Encoder x4	Wheel-speed / motion validation (secondary)





KPR Institute of
Engineering and
Technology
Learn Beyond
(Autonomous, Affiliated to Anna University)



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*Thank
you*

